A counterexample to Oertel's conjecture

Abstract

We construct a rational polytope in $\mathbb{R}^{12} \times \mathbb{R}^5$ of positive mixed-integer volume, the sum of the five-dimensional volumes of its integer fibers, such that every point of $\mathbb{Z}^{12} \times \mathbb{R}^5$ lies in a closed halfspace holding at most a fixed fraction $r < (5/6)^5/2^{12}$ of this volume. This disproves Oertel's conjecture, stated as Conjecture 3.5 by Basu and Oertel, that for $d \geq 1$ every convex body of positive mixed-integer volume in $\mathbb{R}^n \times \mathbb{R}^d$ has a point of $\mathbb{Z}^n \times \mathbb{R}^d$ such that every closed halfspace containing it holds at least the fraction $2^{-n}(d/(d+1))^d$ of this volume. Beyond the fibers isolated by tilted caps, the polytope has light fibers that add to the total volume, while each of the three cuts covering the origin fiber pays for only one of three pools of them. The strict comparisons use exact rational arithmetic, and Lean 4 verifies the uniform bound on normalized halfspace depth.

1 Introduction

Cutting plane methods for convex minimization measure progress by the volume they remove from a search region, so a query point with a large fraction of the volume in every closed halfspace containing it gives control over the progress of a cut. Over $\mathbb{R}^d$ such a point always exists. Grünbaum [5, Theorem 2 and Section 3] proved that every closed halfspace containing the centroid of a convex body in $\mathbb{R}^d$, $d \geq 1$, holds at least the fraction $\Gamma_d = (d/(d+1))^d$ of its volume, and this fraction is best possible [5, Section 4]. For convex mixed-integer optimization the volume becomes a sum of fiber volumes, and Basu and Oertel [1, Section 2 and Theorem 2.6] bound the number of first-order oracle calls of a centerpoint cutting plane method through the fraction guaranteed at the query point. We show that the fraction conjectured by Oertel fails at $(n, d) = (12, 5)$.

For nonnegative integers n, d with $n + d > 0$ we write points of $\mathbb{R}^n \times \mathbb{R}^d$ as $(\mathbf{z}, \mathbf{x})$. The *mixed-integer lattice* is $\mathbb{Z}^n \times \mathbb{R}^d$, and the *fiber* of a set $D \subseteq \mathbb{R}^n \times \mathbb{R}^d$ over $\mathbf{z} \in \mathbb{Z}^n$ is $D_{\mathbf{z}} = \{\mathbf{x} \in \mathbb{R}^d : (\mathbf{z}, \mathbf{x}) \in D\}$. The *mixed-integer volume*, or *mass*, of a set with Lebesgue measurable fibers is the sum of the d-dimensional volumes of its fibers. A *proper closed halfspace* is a set $\{F \geq b\} = \{(\mathbf{z}, \mathbf{x}) : F(\mathbf{z}, \mathbf{x}) \geq b\}$, where F is a nonzero linear functional on $\mathbb{R}^n \times \mathbb{R}^d$ and $b \in \mathbb{R}$. For a convex body $C \subset \mathbb{R}^n \times \mathbb{R}^d$, that is, a compact convex set with nonempty interior, of positive mass, the *depth* of a point $(\mathbf{z}, \mathbf{x}) \in \mathbb{Z}^n \times \mathbb{R}^d$ is the infimum of the normalized mass over the proper closed halfspaces containing it. With vol denoting Lebesgue measure on $\mathbb{R}^d$ for $d \geq 1$ and counting measure on the singleton $\mathbb{R}^0$ for $d = 0$,

$$\mu(D) = \sum_{\mathbf{z} \in \mathbb{Z}^n} \operatorname{vol}(D_{\mathbf{z}}), \qquad \operatorname{depth}_C(\mathbf{z}, \mathbf{x}) = \inf_{H \ni (\mathbf{z}, \mathbf{x})} \frac{\mu(C \cap H)}{\mu(C)}.$$

Compactness leaves finitely many nonempty fibers, so $\mu(C) < \infty$. A halfspace containing a point contains the parallel halfspace with that point on its boundary, so the infimum is unchanged if we use only the latter, which is the form in [1, (2)]. Points outside C have depth 0 by strict separation, and we put $\Gamma_0 = 1$.

Two extreme cases suggest the fraction to expect. For $n = 0$ the mass is the volume, and Grünbaum's theorem gives depth at least Γ_d at the centroid. For $d = 0$ the mass counts integer points, and the general bound $1/(2^n(d+1))$ of Oertel [6, Theorem 4.1.19] and Basu and Oertel [1, Corollary 3.4] gives 2^{-n}. The product of the unit cube and a d-simplex combines the two factors [6, Section 4.1, after Conjecture 4.1.20], since its largest depth is exactly $2^{-n}\Gamma_d$ (Section 1.1). Oertel [6, Conjecture 4.1.20], for closed convex sets, and Basu and Oertel [1, Conjecture 3.5], for convex bodies,

conjectured that this fraction is always reached. In particular, every convex body of positive mass would have a point of $\mathbb{Z}^n \times \mathbb{R}^d$ of depth at least $2^{-n}\Gamma_d$, which equals $2^{-n}(d/(d+1))^d$ for $d \geq 1$. We disprove this at $(n, d) = (12, 5)$, where the fraction is $(5/6)^5/2^{12}$.

Theorem 1.1. There is a rational polytope $C \subset \mathbb{R}^{12} \times \mathbb{R}^5$ with nonempty interior, $\mathbf{0} \in C$ and $0 < \mu(C) < \infty$ such that $\sup_{(\mathbf{z},\mathbf{x})\in\mathbb{Z}^{12}\times\mathbb{R}^5} \operatorname{depth}_C(\mathbf{z}, \mathbf{x}) < (5/6)^5/2^{12}$.

The inequality for the supremum gives a uniform margin over the lattice. The proof gives more, namely an explicit number $r = M/W < (5/6)^5/2^{12}$ and, for every point of $\mathbb{Z}^{12} \times \mathbb{R}^5$, a proper closed halfspace containing the point with normalized mass at most r. It compares explicit masses in exact rational arithmetic (Section 5), and Lean 4 verifies this uniform strict bound for a compact convex set (Appendix A). Rationality of C is part of the mathematical proof (Lemma 4.2).

For $d \geq 2$ the general bound $1/(2^n(d+1))$ lies below the conjectured fraction, and the following results go beyond it under additional structure. For large projections onto $\mathbb{R}^n$, Basu and Oertel [1, Theorem 3.6] proved the conjectured bound when the projection has lattice width above a threshold exponential in n, Cristi and Salas [4, Theorem 4.5] proved it when the projection contains a Euclidean ball of radius at least $\alpha d^2 n^{3/2}$ for a universal constant α, and Cheng and Basu [2, Corollary 2.5] showed that a translate of the ℓ_∞ ball of radius $k \geq 3e(n+d)$ in the projection gives depth at least $1/(2e)$. For few fibers, Cheng and Basu [3, Theorem 1] proved the fraction $2/9 = \Gamma_2/2$ for compact convex sets in $\mathbb{R} \times \mathbb{R}^2$ with exactly three slices of positive area.

1.1 From the product to the construction

Let $S \subset \mathbb{R}^d$ be a d-simplex and $C = [0, 1]^n \times S$, so that $\mu(C) = 2^n \operatorname{vol}(S)$. A point $\mathbf{x} \in S$ has a barycentric coordinate at least $1/(d+1)$, and the cap of S where this coordinate is at least $1/(d+1)$ is a homothetic copy of S with ratio $d/(d+1)$, of volume $\Gamma_d \operatorname{vol}(S)$. For $d = 0$ the simplex is a point and the cap is this point. For $\mathbf{z} \in \{0, 1\}^n$, adding a large multiple of a linear functional uniquely maximized over $\{0, 1\}^n$ at $\mathbf{z}$ tilts this cap into a closed halfspace that misses all other fibers, and for $d = 0$ this exposing functional alone gives the cut. Hence every point of $\mathbb{Z}^n \times \mathbb{R}^d$ has depth at most $2^{-n}\Gamma_d$ in this C. Conversely, a proper closed halfspace containing a point $(\mathbf{z}, \mathbf{x})$ restricts on the fiber over $\mathbf{z}$ to a closed halfspace of $\mathbb{R}^d$ containing $\mathbf{x}$ or to all of $\mathbb{R}^d$. So for $d \geq 1$ Grünbaum's theorem on any one fiber gives its centroid depth at least $2^{-n}\Gamma_d$, and for $d = 0$ a halfspace containing a point of C holds its mass $1 = 2^{-n}\mu(C)$. The largest depth is therefore exactly $2^{-n}\Gamma_d$. The same restriction also limits any counterexample. If a convex body has at most 2^n fibers of positive volume, then a largest fiber carries at least 2^{-n} of the mass, and its centroid, or its point when $d = 0$, has depth at least $2^{-n}\Gamma_d$. A counterexample therefore needs more than 2^n such fibers. The polytope of Theorem 1.1 has $4852 > 2^{12}$ of them, and its projection onto $\mathbb{R}^{12}$ lies in $[-1, 1]^{12}$.

Our proof is a mass count against one budget $M > 0$ for all *cuts*, that is, proper closed halfspaces. If every point of the lattice lies in a cut of mass at most M, then its depth is at most M/W with $W = \mu(C)$, and $M/W < (5/6)^5/2^{12}$ is equivalent to $W/M > 2^{12}(6/5)^5$. We write $M = (5h/6)^5$, which is the mass of a tilted cap of a simplex fiber of volume h^5, so the target is $W > 4096h^5$. The polytope C is the convex hull of fibers over 4852 integer points, which we call bases, and we use units where the origin fiber has volume 1.

Three masses add up to W. One *heavy* base lies in each of the 4095 nonzero classes of $(\mathbb{Z}/2\mathbb{Z})^{12}$. It is a vertex of the convex hull of the bases, so every point of its fiber lies in a cap of mass at most M that can be tilted in the integer directions to miss all other fibers. Heavy fibers have volume at most h^5 and the origin fiber has volume $1 < h^5$, so these 4096 fibers alone fall short of the target. The rest comes from 756 *light* bases, each the negative of a heavy base, which the selector

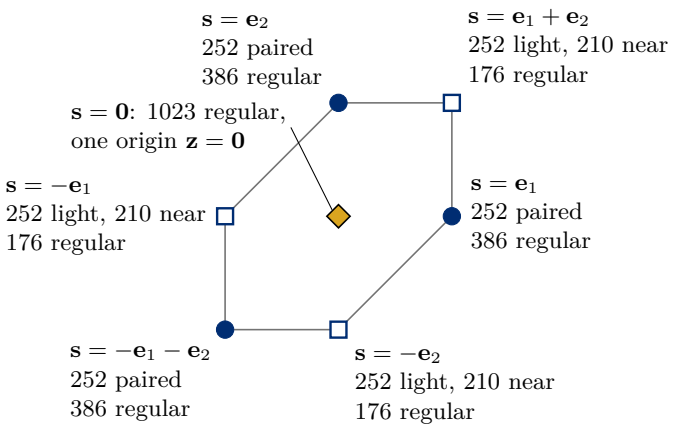


Figure 1: The bases by selector, the last two integer coordinates $\mathbf{s}$, where $\mathbf{e}_1, \mathbf{e}_2$ are the unit vectors of $\mathbb{R}^2$. Each marker is one layer of bases with the same value of $\mathbf{s}$, labelled with its counts, and the hexagon is the convex hull of the seven plotted selectors. Circles mark selectors containing paired bases, the heavy bases with a light partner, squares mark their antipodal selectors containing light bases, and the diamond marks the layer of the origin. Near and regular bases are the other heavy bases (Section 2).

coordinates sort into three pools of 252 (Figure 1). All three pools, each of mass L, add to W, which thus gains $3L$. A light base and its heavy partner have the origin as their midpoint, so a tilt in the integer coordinates alone cannot separate the origin from both. Instead, three coordinated cuts cover the origin fiber, and each has positive mass only on its cap of the origin fiber and on one pool. Each origin cut thus pays for L once, alongside the origin cap, and only this sum must stay below M. The total $3L$ exceeds the amount by which the origin fiber and the heavy fibers with light partners fall short of h^5, which yields the target. To keep the origin fiber unchanged in the convex hull, we place each fiber inside a dilation of the origin fiber whose radius is affine in $\mathbf{z}$ and equals 1 at the origin, so that antipodal bases have radii with sum 2.

Section 2 defines the bases and the three linear forms that offset the origin cuts. Section 3 derives the fibers from the requirements of these cuts, Section 4 proves that C has no other integer fibers, and Section 5 constructs a cut at every point and completes the exact comparison. Appendix A records the Lean statement, the exact floors and the replay.

2 Integer bases and offset forms

Write $\mathbf{z} = (\mathbf{y}, \mathbf{s}) \in \mathbb{R}^{10} \times \mathbb{R}^2$, with ordinary coordinates $\mathbf{y}$ and selector coordinates $\mathbf{s}$, and put $u = \sum_i y_i$. The linear forms

$$\tau(\mathbf{z}) = (u - s_2,\ u - s_1,\ u - s_1 - s_2)$$

will offset the three cuts over the origin, and the sum of their values, $3u - 2s_1 - 2s_2$, will control the radii of the fibers.

Two choices make τ useful. Let $\mathbf{e}_1, \mathbf{e}_2$ be the unit vectors of $\mathbb{R}^2$. The selectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2$ have the three nonzero residues of $(\mathbb{Z}/2\mathbb{Z})^2$, and each will carry a separate copy of the ten-dimensional Boolean cube. Let $c_i = -1$ for $i \le 4$ and $c_i = 1$ for $i \ge 5$. For $J \subseteq \{1, \dots, 10\}$, let $\mathbf{1}_J$ be the indicator vector of J, and let $\mathbf{b}_J$ have entries c_i on J and 0 off J. The sign change does not alter the residue modulo 2, but it moves the ordinary sum. Since $u(\mathbf{b}_J) = |J \triangle \{1, 2, 3, 4\}| - 4$, toggling

membership in the first four positions shows that $u = k$ has $\binom{10}{k+4}$ solutions. In particular, $u = 1$ has $\binom{10}{5} = 252$ solutions, whereas the unsigned masks $\mathbf{1}_J$ have only 10. The *candidates* are $(\mathbf{1}_J, \mathbf{0})$ for $J \neq \emptyset$ and $(\mathbf{b}_J, \mathbf{s})$ for $\mathbf{s} \in \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2\}$, the latter with the following values.

Type j	$\mathbf{s}$	$\tau(\mathbf{b}_J, \mathbf{s})$	Sum of entries	Paired	Near
1	$\mathbf{e}_1$	$(u,\ u-1,\ u-1)$	$3u-2$	$u=1$	$u=0$
2	$\mathbf{e}_2$	$(u-1,\ u,\ u-1)$	$3u-2$	$u=1$	$u=0$
3	$\mathbf{e}_1+\mathbf{e}_2$	$(u-1,\ u-1,\ u-2)$	$3u-4$	$u=1$	$u=2$

The sum of the entries of τ is not divisible by 3 in this table and equals $3|J|$ at $(\mathbf{1}_J, \mathbf{0})$, so it is nonzero on every candidate. A *heavy base* is a candidate multiplied by the sign of this sum. After this orientation, the paired value of u in type j gives $\tau_j = 1$ and zeros elsewhere, the near value gives $\tau_j = 0$ and ones elsewhere, and every other value of u gives three positive entries, as do the bases $(\mathbf{1}_J, \mathbf{0})$. We call these heavy bases *paired* of type j, *near* of type j and *regular*, respectively. For each paired base $\mathbf{z}$ of type j, the point $-\mathbf{z}$ is a *light* base of type j, with $\tau_j = -1$ and zeros elsewhere. The set $\mathcal{Z}$ of *bases* consists of $\mathbf{0}$ and all heavy and light bases.

The orientation is easiest to see on one mask. For $J = \{1, 5, 6\}$, the vector $\mathbf{b}_J$ has entries $-1, 1, 1$ in positions $1, 5, 6$, and $u = 1$. Then $\tau(\mathbf{b}_J, \mathbf{e}_1) = (1, 0, 0)$ and $\tau(\mathbf{b}_J, \mathbf{e}_2) = (0, 1, 0)$, so these two candidates are paired bases as they stand. However, $\tau(\mathbf{b}_J, \mathbf{e}_1 + \mathbf{e}_2) = (0, 0, -1)$ has negative sum, so the paired base of type 3 is $-(\mathbf{b}_J, \mathbf{e}_1 + \mathbf{e}_2)$ with $\tau = (0, 0, 1)$, and its light partner sits at the positive selector. In the fourth parity class of this mask, $(\mathbf{1}_J, \mathbf{0})$ is a regular base with $\tau = (3, 3, 3)$. Figure 2 shows these seven bases and, for reference, the origin.

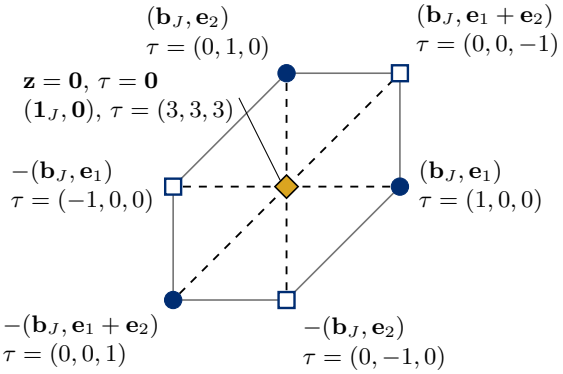


Figure 2: The bases of the mask $J = \{1, 5, 6\}$ at the layers of Figure 1, with the same marks. Each label gives the full base, including the sign of the ordinary part, and its value of τ. A dashed segment joins antipodal bases, whose midpoint is the origin $\mathbf{z} = \mathbf{0}$. The base $(\mathbf{1}_J, \mathbf{0})$ has the same selector as the origin but is a different point of $\mathbb{Z}^{12}$.

Sorting all bases by selector gives Figure 1. For types 1 and 2 the sum is positive exactly when $u \geq 1$, so the heavy base lies at selector $\mathbf{s}$ for $u \geq 1$ and at $-\mathbf{s}$ for $u \leq 0$. For type 3 the sum is positive exactly when $u \geq 2$, so the heavy base lies at $\mathbf{e}_1 + \mathbf{e}_2$ for $u \geq 2$ and at $-(\mathbf{e}_1 + \mathbf{e}_2)$ for $u \leq 1$. Summing $\binom{10}{u+4}$ gives $252 + 386$ paired and regular bases at each of $\mathbf{e}_1, \mathbf{e}_2, -(\mathbf{e}_1 + \mathbf{e}_2)$, and $210 + 176$ near and regular bases at each of their antipodes, which also hold 252 light bases.

Modulo 2, an oriented $(\mathbf{b}_J, \mathbf{s})$ reduces to $(\mathbf{1}_J, \mathbf{s})$, and the three selectors have distinct nonzero residues. Hence the heavy bases represent each nonzero class of $(\mathbb{Z}/2\mathbb{Z})^{12}$ exactly once, and there are 4095 of them, namely 756 paired, 630 near and $3 \cdot 386 + 3 \cdot 176 + 1023 = 2709$ regular. A light base $-\mathbf{z}$ has its partner's residue and differs from it, so the 4852 points of $\mathcal{Z}$ are distinct.

3 The simplex, the radii and the fibers

Let $B = \{\mathbf{x} \in \mathbb{R}^6 : x_i \geq 0,\ \sum_i x_i = 1\}$ be the standard simplex with vertices $\mathbf{v}_1, \ldots, \mathbf{v}_6$. We identify the hyperplane $\sum_i x_i = 1$ with $\mathbb{R}^5$ by dropping x_6 and subtracting $1/6$ from the other coordinates. The centroid of B is then the origin of $\mathbb{R}^5$, scalar multiples of subsets are dilations about the centroid, and $x_1, \ldots, x_6$ are affine functions on $\mathbb{R}^5$ with sum 1. From now on vol and μ are measured in units of the Lebesgue volume of B, which does not change the ratios in Theorem 1.1.

We introduce the parameters of the construction with the conditions they must satisfy. Their exact values are fixed in Section 5, and until then only these conditions are used. Fix a rational number a with $1/2 < a < 1$. For $j = 1, 2, 3$, with these three indices taken cyclically, put $f_j = x_j - a x_{j+1} - (1-a) x_{j+2}$. Each f_j vanishes at the centroid and equals 1 at $\mathbf{v}_j$, so it is a nonzero linear functional on $\mathbb{R}^5$. Moreover $|f_j| \leq 1$ on B and $f_1 + f_2 + f_3 = 0$. The cyclic permutation of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ is a volume-preserving linear symmetry of B that carries f_j to f_{j+1}, so we describe the sets below for $j = 1$ and obtain the others by rotation. The three cuts over the origin are the proper closed halfspaces

$$H_j = \{(\mathbf{z}, \mathbf{x}) \in \mathbb{R}^{12} \times \mathbb{R}^5 : f_j(\mathbf{x}) \geq 3\tau_j(\mathbf{z})\}, \qquad j = 1, 2, 3.$$

Over $\mathbf{z} = \mathbf{0}$ their traces are $B \cap \{f_j \geq 0\}$, and these cover B because $f_1 + f_2 + f_3 = 0$. At another base, the value of τ_j decides what the fiber may look like. If $\tau_j(\mathbf{z}) \geq 1$, then H_j misses a fiber on which $f_j < 3$. If $\tau_j(\mathbf{z}) = 0$, then the trace of H_j on a fiber contained in $\{f_j \leq 0\}$ lies in the hyperplane $\{f_j = 0\}$, which is null. If $\tau_j(\mathbf{z}) = -1$, then H_j contains a fiber on which $f_j \geq -3$. So, by Section 2, a paired or light fiber of type j should lie in $\{f_i \leq 0\}$ for both $i \neq j$, a near fiber of type j should lie in $\{f_j \leq 0\}$, and a regular fiber needs no sign condition. These conditions pass through dilations about the origin because the f_j are linear.

The convex hull of the fibers must also leave the fiber over the origin unchanged. Fix a rational number δ with $0 < \delta < 1$ and put $g(\mathbf{z}) = \delta(\tau_1(\mathbf{z}) + \tau_2(\mathbf{z}) + \tau_3(\mathbf{z})) = \delta(3u - 2s_1 - 2s_2)$. We will place each fiber over $\mathbf{z}$ inside $(1 + g(\mathbf{z}))B$. Since g is linear, antipodal bases get the radii $1 + g$ and $1 - g$, and Lemma 4.2 shows that the origin fiber stays B under arbitrary convex combinations. This is the design we use, and it is not forced for fibers of other shapes. Here g equals δ at paired bases, $-\delta$ at light bases, 2δ at near bases and at least 3δ at regular bases.

Each f_j is a homogeneous linear form in (x_1, x_2, x_3). Non-strict inequalities $f_j \geq 0$ or $f_j \leq 0$ therefore cut out a join of their trace on the active face $\operatorname{conv}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ with $\operatorname{conv}(\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6)$. Indeed, normalizing the first three coordinates by their sum when it is positive preserves the inequalities, and points with that sum zero belong to the latter triangle. The simplex spanned by three points of the active face and $\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6$ has volume equal to the absolute determinant of the 3×3 matrix of barycentric coordinates of these points, because the affine map carrying the vertices of B to the vertices of this simplex acts on barycentric coordinates as the direct sum of this matrix and the identity. We call this the determinant formula. On the active face, f_1 is positive at $\mathbf{v}_1$, negative at $\mathbf{v}_2$ and $\mathbf{v}_3$, and zero at $(a\mathbf{v}_1 + \mathbf{v}_2)/(1+a)$ and $((1-a)\mathbf{v}_1 + \mathbf{v}_3)/(2-a)$. So $B \cap \{f_1 \geq 0\}$ is a simplex, and the determinant formula gives its volume $\beta = 1/[(1+a)(2-a)]$.

The sign conditions and the radius confine a paired or light fiber of type j to a dilation of $T_j = B \cap \{f_{j+1} \leq 0,\ f_{j+2} \leq 0\}$, on which $f_j = -f_{j+1} - f_{j+2} \geq 0$. Among the vertices of the active face, f_2 is positive only at $\mathbf{v}_2$ and f_3 only at $\mathbf{v}_3$. The lines $f_2 = 0$ and $f_3 = 0$ meet at the centroid $\bar{\mathbf{v}} = (\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3)/3$ of the face. They meet the edges from $\mathbf{v}_1$ to $\mathbf{v}_2$ and from $\mathbf{v}_1$ to $\mathbf{v}_3$ at $\mathbf{w}_2 = (\mathbf{v}_1 + (1-a)\mathbf{v}_2)/(2-a)$ and $\mathbf{w}_3 = (\mathbf{v}_1 + a\mathbf{v}_3)/(1+a)$, respectively. The edge from $\mathbf{v}_2$ to $\mathbf{v}_3$ has $f_1 < 0$ throughout, so it is disjoint from T_1. The trace of T_1 is the quadrilateral with vertices $\mathbf{v}_1, \mathbf{w}_2, \bar{\mathbf{v}}, \mathbf{w}_3$ in Figure 3(a), whose diagonal from $\mathbf{v}_1$ to $\bar{\mathbf{v}}$ lies on the line $x_2 = x_3$. Consequently

$T_1 = U \cup P$, where

$$U = \mathrm{conv}(\mathbf{v}_1, \bar{\mathbf{v}}, \mathbf{w}_3, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6), \qquad P = \mathrm{conv}(\mathbf{v}_1, \bar{\mathbf{v}}, \mathbf{w}_2, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6).$$

These simplices lie on opposite sides of the hyperplane $x_2 = x_3$ and meet in a common facet, which is a null set. By the determinant formula, U has volume $\theta = a/[3(1+a)]$ and P has volume $\phi = (1-a)/[3(2-a)]$.

A light fiber can be all of $(1-\delta)T_j$, because the only cut we use on it is H_j, whose mass is computed as a whole. A paired fiber needs caps of mass at most M. We start from the simplex $(1+\delta)U$ because a simplex has an explicit cap bound. At our parameters its volume falls short of h^5, and the total deficit over the 756 paired bases exceeds the light mass. We therefore add a thin slab of P along the common facet, which restores most of the volume and keeps the cap calculation tractable. Section 5 gives the exact volumes.

To define the slab, let $\nu_0, \dots, \nu_5$ be the barycentric coordinates of U in the listed vertex order and put $q_i = \nu_i(\mathbf{w}_2)$. Comparing active coordinates gives $q_1 = 3(1-a)/(2-a)$, $q_2 = -(1-a^2)/[a(2-a)]$, $q_0 = 1 - q_1 - q_2$ and $q_3 = q_4 = q_5 = 0$. Since $1/2 < a < 1$, we have $0 < q_1 < 1$, $q_2 < 0$ and $0 < q_1 + q_2 = q_1(2a-1)/(3a) < 1$, so $0 < q_0 < 1$. Fix a rational number ε with $0 < \varepsilon < 1/1000$ and put $K_1 = T_1 \cap \{\nu_2 \geq \varepsilon q_2\}$. On U we have $\nu_2 \geq 0 > \varepsilon q_2$. On P, let p be the barycentric coordinate at $\mathbf{w}_2$ and, for $i \neq 2$, let p_i be the barycentric coordinate at the vertex of U with index i. Then $\nu_i = p_i + q_i p$ for $i \neq 2$ and $\nu_2 = q_2 p$, since both sides are affine and agree at the vertices of P. Therefore K_1 is the union of U and $P \cap \{p \leq \varepsilon\}$. The set $P \cap \{p \geq \varepsilon\}$ is a homothetic copy of P with ratio $1-\varepsilon$, so $\mathrm{vol}(K_1) = V := \theta + \phi[1-(1-\varepsilon)^5]$. Rotation gives K_2, K_3 and their coordinates ν_i. The inequality $\nu_2 \geq \varepsilon q_2$ has a nonzero constant term in the homogeneous coordinates (x_1, x_2, x_3), so K_1, unlike T_1, U, P, is not a join of its active trace with $\mathrm{conv}(\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6)$. Figure 3(b) shows only that trace.

A near base of type j has radius $1+2\delta$ and needs a fiber in $\{f_j \leq 0\}$. Fix a rational number $h > 0$ with $2h < 1+2\delta$ and put $\eta = (h/(1+2\delta))^5$. Then $\eta < 1/32$, and $a > 1/2$ gives $a/(1+a) > 1/3$, so $\eta < a/(1+a)$. Let $N_j = \mathrm{conv}\big(\eta\mathbf{v}_j + (1-\eta)\mathbf{v}_{j+1}, \mathbf{v}_{j+1}, \mathbf{v}_{j+2}, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6\big) \subseteq B$. At these vertices f_j takes the values $\eta - a(1-\eta)$, $-a$, $a-1$, 0, 0, 0, which are nonpositive because $\eta < a/(1+a)$. Hence $f_j \leq 0$ on N_j, and $\mathrm{vol}(N_j) = \eta$ by the determinant formula. We assign the following fibers to the bases.

Base $\mathbf{z}$	Number	$g(\mathbf{z})$	$\tau(\mathbf{z})$	Fiber $A_{\mathbf{z}}$	$\mathrm{vol}(A_{\mathbf{z}})$
$\mathbf{0}$	1	0	$\mathbf{0}$	B	1
paired, type j	252 per type	δ	$\tau_j = 1$, others 0	$(1+\delta)K_j$	$(1+\delta)^5 V$
light, type j	252 per type	$-\delta$	$\tau_j = -1$, others 0	$(1-\delta)T_j$	$(1-\delta)^5(\theta+\phi)$
near, type j	210 per type	2δ	$\tau_j = 0$, others 1	$(1+2\delta)N_j$	h^5
regular	2709	$\geq 3\delta$	all entries ≥ 1	hB	h^5

The volume of a near fiber is $(1+2\delta)^5\eta = h^5$. Every fiber lies in $(1+g(\mathbf{z}))B$, and the radius $1+g(\mathbf{z})$ is positive because $\delta < 1$. For regular bases this containment uses $h < 1+2\delta < 1+g(\mathbf{z})$ and $\mathbf{0} \in B$. Every heavy fiber lies in $(1+2\delta)B$, so $|f_j| \leq 1+2\delta < 3$ on it because $\delta < 1$. Let $C = \mathrm{conv}\big(\bigcup_{\mathbf{z}\in\mathcal{Z}} \{\mathbf{z}\} \times A_{\mathbf{z}}\big)$. The light fibers of one type have total volume $L = 252(\theta+\phi)(1-\delta)^5$, and all fibers have total volume $W = 3339h^5 + 756(1+\delta)^5 V + 1 + 3L$. The comparison with the target of Section 1 therefore reads

$$W - 4096h^5 = (1-h^5) + 756\big[(1+\delta)^5 V - h^5\big] + 3L. \tag{1}$$

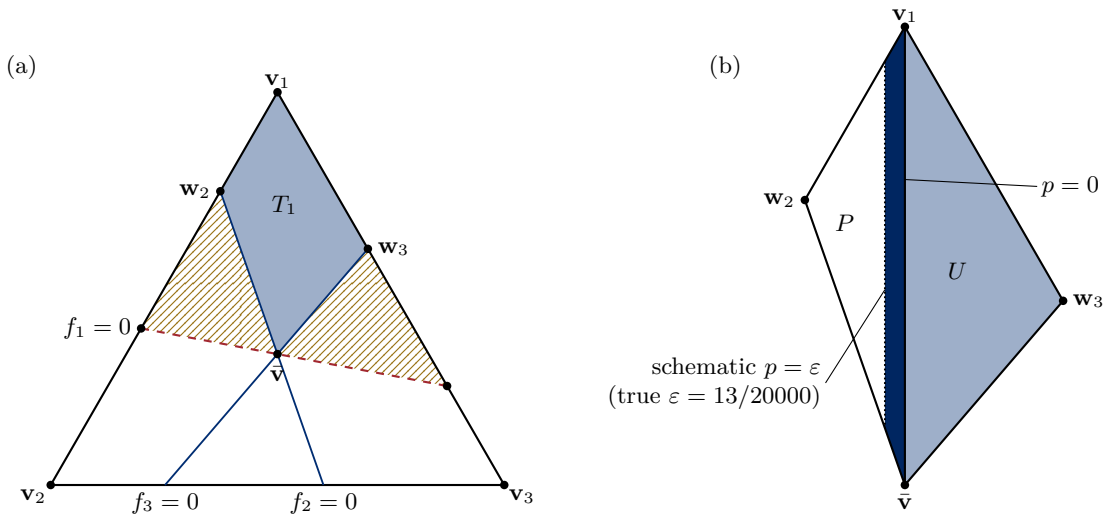


Figure 3: Traces on the active face $x_4 = x_5 = x_6 = 0$, drawn at $a = 663/1000$. These are planar traces and do not show five-dimensional volumes. The point $\bar{\mathbf{v}}$ is the centroid of this face, and the centroid of B, about which all fibers are dilated, lies off this face. (a) The triangle above the dashed line $f_1 = 0$ is the trace of the cap $B \cap \{f_1 \geq 0\}$. Inside it, the shaded quadrilateral is the trace of T_1, and hatching marks the rest of the cap. (b) The trace of T_1. The diagonal $p = 0$ separates U from P, and the two shaded regions together form the trace of K_1. The dark strip is the trace of the five-dimensional slab $P \cap \{p \leq \varepsilon\}$, and its width is exaggerated, drawn at $p = 1/5$ for the true $\varepsilon = 13/20000$.

4 The integer fibers of C

We first show that a nonzero integer point of $\mathrm{conv}(\mathcal{Z})$ has only the trivial representation.

Lemma 4.1. If $\mathbf{z} \in \mathbb{Z}^{12} \setminus \{\mathbf{0}\}$ is a convex combination $\sum_m \lambda_m \mathbf{z}^m$ of points $\mathbf{z}^m \in \mathcal{Z}$ with all $\lambda_m > 0$, then $\mathbf{z}^m = \mathbf{z}$ for every m.

Proof. All entries of all bases lie in $\{-1, 0, 1\}$, so $\mathbf{z} = (\mathbf{y}, \mathbf{s})$ has the same property. If a coordinate of $\mathbf{z}$ equals ± 1, or if the contributors $\mathbf{z}^m = (\mathbf{y}^m, \mathbf{s}^m)$ take only two consecutive integer values in that coordinate, then integrality forces all contributors to agree in it. We call this saturation.

Suppose first that $\mathbf{s} \neq \mathbf{0}$, say $s_k = \pm 1$. Saturation gives $s_k^m = s_k$ for all m. Every nonzero selector of a base is $\pm\mathbf{e}_1$, $\pm\mathbf{e}_2$ or $\pm(\mathbf{e}_1 + \mathbf{e}_2)$, so the other selector coordinate of each contributor lies in $\{0, s_k\}$, and saturation gives $\mathbf{s}^m = \mathbf{s}$. Let $\omega = \pm 1$ be the sign of the nonzero entries of $\mathbf{s}$. By construction, a base with this selector has ordinary part $\omega \mathbf{b}_J$, so $y_i^m \in \{0, \omega c_i\}$ for each i, and saturation gives $\mathbf{y}^m = \mathbf{y}$.

Now suppose that $\mathbf{s} = \mathbf{0}$. Then $y_i = \pm 1$ for some i, and saturation gives $y_i^m = y_i$ for all m. A contributor with a nonzero selector of sign ω has $y_i = \omega c_i$, so $c_i y_i (s_1^m + s_2^m) = \omega(s_1^m + s_2^m) > 0$, while a contributor with selector $\mathbf{0}$ gives 0. The weighted sum of these numbers is $c_i y_i (s_1 + s_2) = 0$, so all contributors have selector $\mathbf{0}$. Their ordinary parts lie in $\{0, 1\}^{10}$, and saturation gives $\mathbf{y}^m = \mathbf{y}$. $\square$

Consequently $\mathbb{Z}^{12} \cap \mathrm{conv}(\mathcal{Z}) = \mathcal{Z}$, and no nonzero base lies in the convex hull of the other bases. For a point $\mathbf{d}$ outside a compact convex set, let $\mathbf{d}_0$ be a nearest point in the set. Minimizing the squared distance to $\mathbf{d}$ on segments from $\mathbf{d}_0$ gives $\langle \mathbf{d} - \mathbf{d}_0, \mathbf{d}' - \mathbf{d}_0 \rangle \leq 0$ for every point $\mathbf{d}'$ of the set. The inner product with $\mathbf{d} - \mathbf{d}_0$ is thus strictly larger at $\mathbf{d}$ than on the set. In particular, each nonzero $\mathbf{z} \in \mathcal{Z}$ has a linear functional $\ell_{\mathbf{z}}$ on $\mathbb{R}^{12}$ with $\ell_{\mathbf{z}}(\mathbf{z}') < \ell_{\mathbf{z}}(\mathbf{z})$ for all $\mathbf{z}' \in \mathcal{Z} \setminus \{\mathbf{z}\}$.

Lemma 4.2. The set C is a rational polytope with nonempty interior and $\mathbf{0} \in C$. For $\mathbf{z} \in \mathbb{Z}^{12}$, the slice $\{\mathbf{x} : (\mathbf{z}, \mathbf{x}) \in C\}$ equals $A_{\mathbf{z}}$ if $\mathbf{z} \in \mathcal{Z}$ and is empty otherwise. Consequently $\mu(C) = W$, and every closed halfspace H satisfies $\mu(C \cap H) = \sum_{\mathbf{z} \in \mathcal{Z}} \operatorname{vol}\big(A_{\mathbf{z}} \cap \{\mathbf{x} : (\mathbf{z}, \mathbf{x}) \in H\}\big)$.

Proof. The fibers are polytopes with rational vertices, because T_j and K_j are cut from B by rational inequalities, N_j has rational vertices, and the dilation factors are rational. So C is the convex hull of finitely many rational points. It contains $\{\mathbf{0}\} \times B$, hence the origin. The set B spans the continuous directions. The regular bases $(\mathbf{1}_{\{i\}}, \mathbf{0})$ and the near bases $(\mathbf{0}, -\mathbf{e}_1)$ and $(\mathbf{0}, -\mathbf{e}_2)$ span the integer directions, so C has nonempty interior.

Let $(\mathbf{z}, \mathbf{x}) \in C$ with $\mathbf{z}$ integer, and write $(\mathbf{z}, \mathbf{x}) = \sum_m \lambda_m (\mathbf{z}^m, \mathbf{x}^m)$ with $\lambda_m > 0$, $\sum_m \lambda_m = 1$, $\mathbf{z}^m \in \mathcal{Z}$ and $\mathbf{x}^m \in A_{\mathbf{z}^m}$. If $\mathbf{z} \neq \mathbf{0}$, then Lemma 4.1 gives $\mathbf{z}^m = \mathbf{z}$ for all m, so $\mathbf{z} \in \mathcal{Z}$ and $\mathbf{x} \in A_{\mathbf{z}}$ by convexity of $A_{\mathbf{z}}$. If $\mathbf{z} = \mathbf{0}$, then $\mathbf{x}^m/(1+g(\mathbf{z}^m)) \in B$ for all m. Since g is linear, the positive weights $\lambda_m(1 + g(\mathbf{z}^m))$ sum to $1 + g(\mathbf{z}) = 1$. Thus $\mathbf{x}$ is a convex combination of the points $\mathbf{x}^m/(1 + g(\mathbf{z}^m))$ of B with these weights, which gives $\mathbf{x} \in B = A_{\mathbf{0}}$. The reverse inclusions are immediate, and the formulas for μ follow by summing over $\mathcal{Z}$. □

5 Cuts and the exact comparison

We now construct the halfspaces, starting with the three cuts over the origin, which contain the light fibers.

Lemma 5.1. For $j = 1, 2, 3$ we have $\mu(C \cap H_j) = \beta + L$. Each point $(\mathbf{0}, \mathbf{x})$ with $\mathbf{x} \in B$ and each point of a light fiber lies in some H_j.

Proof. We read the trace of H_j on each fiber from the table of Section 3, using Lemma 4.2. Over the origin the trace is $B \cap \{f_j \geq 0\}$, of volume β. If $\tau_j(\mathbf{z}) = -1$, the condition $f_j \geq -3$ holds on the whole light fiber of type j, and these fibers have total volume L. If $\mathbf{z} \neq \mathbf{0}$ and $\tau_j(\mathbf{z}) = 0$, then $\mathbf{z}$ is a paired or light base of a type $k \neq j$, or a near base of type j. Its fiber lies in $\{f_j \leq 0\}$, so the trace lies in the null hyperplane $\{f_j = 0\}$. If $\tau_j(\mathbf{z}) \geq 1$, then $\mathbf{z}$ is heavy and the condition $f_j \geq 3$ fails on its fiber. Hence $\mu(C \cap H_j) = \beta + L$. Since $f_1 + f_2 + f_3 = 0$, each $\mathbf{x} \in B$ has $f_j(\mathbf{x}) \geq 0$ for some j, so $(\mathbf{0}, \mathbf{x}) \in H_j$. A point of a light fiber of type j lies in H_j. □

Tilting a cap in the integer directions handles the heavy bases.

Lemma 5.2. Let $\mathbf{z} \neq \mathbf{0}$ be a base, let ψ be a nonconstant affine function on $\mathbb{R}^5$, and let $\kappa \in \mathbb{R}$. There are a nonzero linear functional F and $b \in \mathbb{R}$ with $F(\mathbf{z}, \mathbf{x}') \geq b$ if and only if $\psi(\mathbf{x}') \geq \kappa$, and with $\mu(C \cap \{F \geq b\}) = \operatorname{vol}(A_{\mathbf{z}} \cap \{\psi \geq \kappa\})$.

Proof. For $\rho > 0$ put $F(\mathbf{z}', \mathbf{x}') = \psi(\mathbf{x}') - \psi(\mathbf{0}) + \rho \ell_{\mathbf{z}}(\mathbf{z}')$ and $b = \kappa - \psi(\mathbf{0}) + \rho \ell_{\mathbf{z}}(\mathbf{z})$. Then F is a nonzero linear functional, and $F(\mathbf{z}, \mathbf{x}') \geq b$ if and only if $\psi(\mathbf{x}') \geq \kappa$. Since $\mathcal{Z}$ is finite, the fibers are compact and $\ell_{\mathbf{z}}(\mathbf{z}') < \ell_{\mathbf{z}}(\mathbf{z})$ for $\mathbf{z}' \neq \mathbf{z}$, we have $F < b$ on every fiber over $\mathbf{z}' \neq \mathbf{z}$ once ρ is large. Lemma 4.2 gives the mass. □

A regular or near fiber is a simplex of volume h^5. Some barycentric coordinate of a point in it is at least $1/6$. That coordinate is affine and nonconstant, and the cap of the simplex where it is at least $1/6$ is a homothetic copy with ratio $5/6$, so Lemma 5.2 gives a halfspace of mass $(5h/6)^5$ containing the point.

A paired fiber is not a simplex, and we cover it with the coordinates ν_i. Fix positive rational numbers $t_0, \ldots, t_5$ with $\sum_i t_i = 1$ and $1/7 < t_i < 1/5$. Let $\mathbf{z}$ be a paired base of type j and let $\mathbf{x} \in (1 + \delta)K_j$. By rotation we compute with $j = 1$. Since $\sum_i \nu_i = \sum_i t_i = 1$ on all of

$\mathbb{R}^5$, there is an index i with $\nu_i(\mathbf{x}/(1+\delta)) \geq t_i$. The function $\mathbf{x}' \mapsto \nu_i(\mathbf{x}'/(1+\delta))$ is affine and nonconstant, so Lemma 5.2 with $\kappa = t_i$ gives a halfspace containing $(\mathbf{z}, \mathbf{x})$ of mass $(1+\delta)^5 V_i$, where $V_i = \mathrm{vol}(K_1 \cap \{\nu_i \geq t_i\})$.

Lemma 5.3. We have $V_2 = \theta(1-t_2)^5$, and for $i \neq 2$,

$$V_i = \theta(1-t_i)^5 + \phi\,\frac{(1-t_i)^5 - \big(1-t_i-(1-q_i)\varepsilon\big)^5}{1-q_i}.$$

Proof. We split K_1 into U and $P \cap \{p \leq \varepsilon\}$, which overlap in a null set. In U the cap $\{\nu_i \geq t_i\}$ is a homothetic copy of U with ratio $1-t_i$, of volume $\theta(1-t_i)^5$. On P with $i = 2$ we have $\nu_2 = q_2 p \leq 0 < t_2$, so this part is empty. Let $i \neq 2$. For $0 \leq p \leq \varepsilon$, the bounds $0 \leq q_i < 1$, $\varepsilon < 1/1000$ and $1/7 < t_i < 1/5$ give $t_i - q_i p > 0$, $1 - t_i - (1-q_i)p > 0$ and $1 - q_i > 0$. The affine chart with coordinates p and p_k, $k \neq i, 2$, sends P onto the standard simplex of coordinate Lebesgue volume $1/120$, so volume in our units on P is 120ϕ times coordinate Lebesgue measure. Substituting $p_i = 1 - p - \sum_{k \neq i,2} p_k$ turns the condition $\nu_i = p_i + q_i p \geq t_i$ into $\sum_{k \neq i,2} p_k \leq 1 - t_i - (1-q_i)p$, and it forces $p_i \geq t_i - q_i p > 0$. So for each p the four remaining coordinates range over a simplex of coordinate Lebesgue volume $[1 - t_i - (1-q_i)p]^4/24$. Multiplying by 120ϕ and integrating over p, the contribution of P is

$$5\phi \int_0^{\varepsilon} \big[1 - t_i - (1-q_i)p\big]^4\, dp = \phi\,\frac{(1-t_i)^5 - \big(1-t_i-(1-q_i)\varepsilon\big)^5}{1-q_i}. \qquad \square$$

The cuts above reduce the theorem to three strict comparisons of explicit rational numbers,

$$\beta + L < M, \qquad (1+\delta)^5 V_i < M \quad (i = 0, \ldots, 5), \qquad 4096 h^5 < W. \tag{2}$$

The first bounds the origin cuts, the second bounds the caps on the paired fibers, and the third is the target of Section 1.1. We take

$$a = \frac{663}{1000}, \qquad \delta = \frac{664201}{10^6}, \qquad \varepsilon = \frac{13}{20000}, \qquad h = \frac{1111946415}{10^9},$$
$$(t_0, \ldots, t_5) = 10^{-9}(166735371,\ 166735294,\ 166324902,\ 166734811,\ 166734811,\ 166734811).$$

They satisfy all the conditions imposed so far, namely $1/2 < a < 1$, $0 < \delta < 1$, $0 < \varepsilon < 1/1000$, $h > 0$, $2h < 1 + 2\delta$, $\sum_i t_i = 1$ and $1/7 < t_i < 1/5$. In lowest terms, $\theta = 221/1663$, $\phi = 337/4011$, $\beta = 1000000/2223431$, $q_0 = 776569/886431$, $q_1 = 1011/1337$, $q_2 = -560431/886431$ and $5h/6 = 74129761/80000000$. Exact evaluation of the formulas of Sections 3 and 5 gives $M - \beta - L > 10^{-6}$, $M - \max_i (1+\delta)^5 V_i > 10^{-9}$ and $W - 4096h^5 > 4 \cdot 10^{-5}$, which proves (2). Appendix A lists the exact floors and the replay.

The third comparison is the mass count of Section 1.1, and it shows what the slab of Section 3 does. The simplex $(1+\delta)U$ falls short of h^5 by more than $3 \cdot 10^{-3}$, so its total deficit over 756 bases exceeds $3L$, while $(1+\delta)K_j$ falls short by less than $4 \cdot 10^{-7}$. Each paired fiber thus remains slightly smaller than a regular one. In (1), the light mass $3L$ exceeds the sum of the two deficits $h^5 - 1$ and $756[h^5 - (1+\delta)^5 V]$, and the surplus is $W - 4096h^5$.

Proof of Theorem 1.1. Lemma 4.2 gives the stated properties of C and $\mu(C) = W$ in our units, and $0 < W < \infty$ because W is a finite sum of positive volumes. A point $(\mathbf{z}, \mathbf{x}) \in \mathbb{Z}^{12} \times \mathbb{R}^5$ either lies outside C or has $\mathbf{z} \in \mathcal{Z}$ and $\mathbf{x} \in A_{\mathbf{z}}$. All inequalities defining the covers are nonstrict, so points on the boundaries of fibers need no separate treatment. If $(\mathbf{z}, \mathbf{x}) \notin C$, nearest-point separation gives

a nonzero F with $F(\mathbf{z},\mathbf{x}) > F$ on C. With $b = F(\mathbf{z},\mathbf{x})$, the halfspace $\{F \ge b\}$ contains the point and misses C, so it has mass 0. If $\mathbf{z} = \mathbf{0}$ or $\mathbf{z}$ is light, Lemma 5.1 gives a containing H_j of mass $\beta + L < M$. If $\mathbf{z}$ is regular or near, the simplex cap gives mass M. If $\mathbf{z}$ is paired, the cover by the ν_i gives mass $(1+\delta)^5 V_i < M$. So every point of $\mathbb{Z}^{12} \times \mathbb{R}^5$ lies in a proper closed halfspace H with $\mu(C \cap H) \le M$.

Put $r = M/W = (5/6)^5 h^5/W$. Then $\mu(C \cap H) \le r\,\mu(C)$, and $W > 4096h^5$ gives $0 < r < (5/6)^5/2^{12}$. Thus for every $(\mathbf{z},\mathbf{x}) \in \mathbb{Z}^{12} \times \mathbb{R}^5$ there are $b \in \mathbb{R}$ and a nonzero linear functional F on $\mathbb{R}^{12} \times \mathbb{R}^5$ with $F(\mathbf{z},\mathbf{x}) \ge b$ and $\mu(C \cap \{F \ge b\}) \le r\,\mu(C)$. Hence $\mathrm{depth}_C(\mathbf{z},\mathbf{x}) \le r$ for every such point, and the supremum is at most $r < (5/6)^5/2^{12}$. Numerically, $10^5 r = 9.811464096\ldots$ and $10^5(5/6)^5/2^{12} = 9.811464160\ldots$. □

A Lean statement and exact replay

The exact checker and Lean project are tracked in PR #14. The checker and Lean sources for this 17-dimensional result are unchanged since the verification source below.

Verification source	`d368306f28ce808d2b2b773cc025745d835538bf`
Lean toolchain	`leanprover/lean4:v4.34.1`
mathlib revision	`d13f23b723b8a846827a245b89c10fc7d3f11612`

In the namespace `SharpOC.Selector17`, the final declaration reads as follows.

```
theorem sharp_oc_counterexample :
    ∃ C : Set Ambient, IsCompact C ∧ Convex ℝ C ∧ (interior C).Nonempty ∧
      ((0 : IntegerSpace), (0 : E)) ∈ C ∧
      0 < mixedIntegerMass C ∧ mixedIntegerMass C ≠ ⊤ ∧
      ∃ r : ℝ, 0 < r ∧ r < (((5 : ℝ) / 6)^5 / (2 : ℝ)^12) ∧
        ∀ (z : Coordinate → ℤ) (x : E),
          ∃ F : Ambient →L[ℝ] ℝ, ∃ b : ℝ, F ≠ 0 ∧ b ≤ F ((fun j => (z j : ℝ)), x) ∧
            (mixedIntegerMass (C ∩ {p | b ≤ F p})).toReal /
              (mixedIntegerMass C).toReal ≤ r
```

Here `Coordinate` is `Fin 10 ⊕ Fin 2`, `IntegerSpace` is `Coordinate → ℝ`, `E` is `Fin 5 → ℝ` and `Ambient` is `IntegerSpace × E`. For a set `D`, the value `mixedIntegerMass D` is the extended nonnegative sum, over all `w : Coordinate → ℤ`, of the Lebesgue volume of `{x | ((fun k => (w k : ℝ)), x) ∈ D}`, which is $\mu(D)$ in the Lebesgue units of Section 1. Finiteness of the mass makes both `toReal` values faithful, so the number `r` and the halfspaces give the bound of Theorem 1.1. Lemma 4.2 establishes rationality of the polytope.

The following floors come from direct exact evaluation, with the parameters of Section 5, of the formulas of Sections 3 and 5.

Mass	$\lfloor 10^{12} \cdot \text{mass} \rfloor$	Margin	$\lfloor 10^{12} \cdot \text{margin} \rfloor$
h^5	1699884034250	$M - \beta - L$	1262439
M	683145268393	$M - (1+\delta)^5 V_0$	7542
β	449755355574	$M - (1+\delta)^5 V_1$	6288
L	233388650379	$M - (1+\delta)^5 V_2$	1211
$3L$	700165951139	$M - (1+\delta)^5 V_i,\ i = 3,4,5$	7869
$(1+\delta)^5\theta$	1696402554056	$h^5 - (1+\delta)^5 V$	312774
$(1+\delta)^5 V$	1699883721475	$1 + 3L - h^5$	281916889
W	6962725049747587	$W - 4096h^5$	45459473

With Python 3, Git and elan installed, run the following from the repository root. Lake reads its dependency pins from `lean/lake-manifest.json`.

```
set -eu
python3 -B research/sharp-dimension-17/check.py
cd lean
lake exe cache get
lake build
lake env lean CheckAxioms.lean
lake env lean --stdin <<'LEAN'
import SharpOC.Selector17Counterexample

#check SharpOC.Selector17.integer_projection_census
#check SharpOC.Selector17.ambient_dimension
#check SharpOC.Selector17.sharp_oc_counterexample
#print axioms SharpOC.Selector17.sharp_oc_counterexample
LEAN
```

The checker uses exact rational arithmetic to generate the 4852 listed bases and their counts. It also asserts the containment and cap inequalities together with the three margins of Section 5, which imply (2). The checker does not show that the list exhausts the integer projection. That statement is proved in Section 4 and in Lean by `integer_projection_census`. The Lean build passed, and this lemma, the dimension lemma and the final theorem depend only on `propext`, `Classical.choice` and `Quot.sound`.

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